曹家彦



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自我评价

【设计能力与研发协同】5 年产品设计与研发经验，涉及工业设计、UI/UX 及软硬件协同设计。能够结合产品功能、用户使用场景与工程约束开展设计工作，将需求转化为产品方案与交互原型，并推动设计落地。

【跨部门协作与落地】能够与结构、软件等团队沟通设计需求、协调实现方案，并跟进设计验证与迭代。熟悉医疗器械产品开发流程，具备技术文档维护及注册申报相关技术资料支持经验。

【AI 辅助原型开发】具备较强的自主学习能力，能够运用 AI 编程工具将交互设计方案转化为高保真、可运行原型，支持设计验证与研发协作，提升原型开发效率。

教育背景

- **美国加州大学洛杉矶分校（UCLA）** | 建筑与城市设计 · 硕士 **2020 年 8 月-2021 年 6 月**
  GPA 3.871/4.0 | 获得校奖学金
- **美国爱荷华州立大学（ISU）** | 建筑学 · 本科（辅修可持续发展） **2015 年 8 月-2020 年 5 月**

资质与知识产权

- **NPDP 新产品开发专业人士认证证书** | 国际产品管理协会 (PDMA) **2025 年 12 月**
- **知识产权成果：6** 项国家专利（**3** 项外观设计专利、**3** 项实用新型专利）及 **1** 项软件著作权（主要完成人）

工作经历

海易特医疗科技（上海）有限公司 | 设计经理 **2021 年 8 月 – 2026 年 8 月**

【产品设计与原型验证】主导/参与 **15+** 款康复产品研发设计，涉及 B 端与 C 端使用场景分析、工业设计与交互设计优化，以及原型迭代与验证；结合产品功能、用户需求与工程约束，推进设计方案落地。

【设计开发文档与合规支持】负责 **20+** 份设计开发文档的编写与维护，参与 NMPA 注册申报相关技术资料的整理与完善，并维护文档的版本一致性与可追溯性。

项目经历

核心项目：

- **视听觉节律刺激步态机器人 【VRRAS】** **2021 年 8 月 – 2026 年 8 月**

利用 Unity 搭建 **5+** 种游戏化康复训练场景；主导整机工业设计与人机工效优化，通过设计方案调整使设备估算重量降低约 **30%**；通过 **7+** 轮实物打样验证束带设计与安全性。

- **下肢康复运动器及其版本迭代 【D1, D2】** **2021 年 8 月 – 2026 年 8 月**

围绕治疗师与患者的不同使用需求开展 **12+** 次交互迭代；负责 D2 版本交互逻辑定义，并借助 AI 编程工具将设计转化为高保真、可运行原型，支持研发对接与设计验证；作为软件著作权主要完成人，参与完成相关登记；协助完成 D1 版 I 类医疗器械注册。

- **步态训练与评估系统 【KS】** **2021 年 9 月 – 2024 年 12 月**

主导系统交互流程设计，通过 **10+** 次界面迭代优化训练引导与操作逻辑，提升操作流程的清晰度；推动防护方案与 VRRAS 的通用化设计，提升体验一致性并减少重复开发。

其他项目：

- **康复 SaaS 平台** **2025 年 3 月 – 2026 年 8 月**

主导前期体验定义，通过内部多角色访谈梳理业务痛点，构建用户画像与用户旅程图，并明确核心功能优先级。

- **手功能康复机器人【GG】** **2023 年 10 月 – 2024 年 2 月**

基于人体尺寸研究推进软织物手套 **5+** 轮打样迭代；参与传感器集成方案设计，兼顾产品可制造性与可维护性。

专业技能

- **设计与原型工具：**Rhinoceros、AutoCAD、KeyShot、V-Ray、Figma、Adobe Suite（PS/AI/ID）、Unity。
- **AI 辅助开发：**Codex、Antigravity、Cursor 等 AI 编程工具、AI 辅助前端原型开发、Midjourney。



Jiayan Cao

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SUMMARY

- **Design & R&D Collaboration: 5** years of experience in product design and development, covering industrial design, UI/UX, and hardware-software integration. Able to translate product requirements into design solutions and interactive prototypes while considering product functionality, user needs, and engineering constraints.
- **Cross-functional Collaboration & Implementation:** Experienced in communicating design requirements, coordinating solutions, and supporting design validation and iteration with cross-functional teams. Familiar with medical device development processes, including technical documentation and regulatory submission support.
- **AI-assisted Prototyping**: Quick to learn new tools and technologies. Experienced in using AI coding tools to turn interaction design concepts into high-fidelity, working prototypes for design validation and collaboration with development teams, improving prototyping efficiency.

EDUCATION

University of California, Los Angeles (UCLA) | Master of Architecture and Urban Design ***Aug. 2020 – Jun. 2021***

- GPA 3.871/4.0 | UCLA M.S.AUD Program Scholarship Recipient

Iowa State University (ISU) | Bachelor of Architecture (Minor in Sustainability) ***Aug. 2015 - May. 2020***

CERTIFICATIONS & INTELLECTUAL PROPERTY

- **NPDP (New Product Development Professional)** | PDMA ***Dec. 2025***
- **IP Assets: 6** Chinese patents (**3** design and **3** utility model patents) and **1** software copyright (primary contributor).

WORK EXPERIENCE

HyACT Medical Technology (Shanghai) Co., Ltd. | Design Manager ***Aug. 2021 – Aug. 2026***

- **Product Design & Prototyping:** Led and contributed to the design and development of **15+** rehabilitation products for B2B and B2C use cases, covering industrial design, interaction design, and prototype testing and iteration. Developed solutions based on product functionality, user needs, and engineering constraints.
- **Design Documentation & Regulatory Support:** Prepared and maintained **20+** design and development documents, including technical documentation and regulatory submission materials, ensuring consistency and traceability.

PROJECT EXPERIENCE

Core Projects:

- **Visual and Auditory Rhythmic Stimulation Gait Rehabilitation Robot (VRRAS)** ***Aug. 2021 – Aug. 2026***

Developed **5+** gamified rehabilitation training scenarios in Unity. Led industrial design and ergonomic improvements for the device, reducing its estimated weight by approximately **30%** through design modifications. Conducted **7+** rounds of physical prototyping to validate the design and safety of the restraint straps.

- **Lower Limb Rehabilitation Exercise Device (D1, D2)** ***Aug. 2021 – Aug. 2026***

Completed **12+** interaction design iterations based on the needs of therapists and patients. Defined the interaction logic for D2 and used AI coding tools to turn interaction designs into high-fidelity, working prototypes for development handoff and design validation. Served as a primary contributor to the software copyright registration. Supported the Class I medical device registration process for D1.

- **Gait Training and Assessment System (KS)** ***Sep. 2021 – Dec. 2024***

Led the design of system interaction flows and completed **10+** UI iterations to improve training guidance and operational clarity. Helped standardize protective design solutions across KS and VRRAS, improving consistency and reducing duplicated development work.

Additional Projects:

- **Rehabilitation SaaS Platform** ***Mar. 2025 – Aug. 2026***

Led early-stage experience definition by interviewing internal stakeholders across different roles to identify business pain points, develop user personas and journey maps, and prioritize core features.

- **Hand Rehabilitation Robot (GG)** ***Oct. 2023 – Feb. 2024***

Conducted anthropometric research and led **5+** rounds of prototyping for a soft-fabric rehabilitation glove. Contributed to a modular sensor integration design, considering both manufacturability and future maintenance.

SKILLS

- **Design & Prototyping Tools:** Rhinoceros, AutoCAD, KeyShot, V-Ray, Figma, Adobe Suite (PS/AI/ID), Unity.
- **AI-assisted Development:** Codex, Antigravity, Cursor, AI-assisted frontend prototyping, Midjourney.